

## Coding & Solution

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 03 July 2026                                                |
| <b>Team ID</b>       | XXXXXX                                                      |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 5 Marks                                                     |

### Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

#### Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

## Solution Summary

| Field                                | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Repository Link / URL</b>         | <a href="https://github.com/varalakshmivennapusa124-cell/OptiCrop.git">https://github.com/varalakshmivennapusa124-cell/OptiCrop.git</a>                                                                                                                                                                                                                                                                                                                                                        |
| <b>Programming Language(s)</b>       | Python, HTML5, CSS3, JavaScript                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Framework(s) Used</b>             | Flask, Scikit-learn, Pandas, NumPy, Matplotlib                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Key Features Implemented</b>      | <ul style="list-style-type: none"> <li>• Soil and environmental data input (N, P, K, Temperature, Humidity, pH, and Rainfall).</li> <li>• Machine learning–based crop recommendation using agricultural data.</li> <li>• Crop suitability analysis with suitability percentage and parameter gap analysis.</li> <li>• Top 3 crop recommendations with confidence scores and farming tips.</li> <li>• Responsive web interface with Flask API integration for real-time predictions.</li> </ul> |
| <b>Pending / Incomplete Features</b> | <ul style="list-style-type: none"> <li>• Real-time weather API integration</li> <li>• Fertilizer recommendation module</li> <li>• Crop disease prediction</li> <li>• Mobile application support</li> </ul>                                                                                                                                                                                                                                                                                     |
| <b>Setup / Run Instructions</b>      | Install the required dependencies, run the Flask backend (python app.py), launch the React application (or open the deployed Vercel URL), and enter soil and environmental details to use the crop recommendation and crop suitability analysis features.                                                                                                                                                                                                                                      |

### Code Quality Checklist

| S.No | Criteria                                               | Status |
|------|--------------------------------------------------------|--------|
| 1    | Code is modular and organized into functions / classes | Yes    |
| 2    | Meaningful variable and function names are used        | Yes    |
| 3    | Code includes comments / documentation where necessary | Yes    |
| 4    | Error handling is implemented for critical operations  | Yes    |
| 5    | The application runs without critical errors           | Yes    |
| 6    | Code is committed to a version control repository      | Yes    |

### Additional Notes / Comments:

The OptiCrop project is designed to help farmers make better crop selection decisions by analyzing soil nutrients and environmental conditions. The application follows a simple and organized structure, making it easy to understand, maintain, and improve. Machine learning is used to generate crop recommendations, while Flask connects the prediction model with a user-friendly web interface. The project can be extended in the future by adding live weather integration, fertilizer recommendations, and crop disease detection.